INCH-POUND

MIL-P-80026C
22 August 1988
SUPERSEDING
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14 December 1976

MILITARY SPECIFICATION

PUNCHING MACHINE, METAL, POWER DRIVEN, TURRET TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers power driven metal punching machines of the multi-station turret type, complete with equipment and accessories specified herein.

1.2 Classification. The punching machine shall be of the following sizes. The size to be furnished shall be as specified (see 6.2.1). Size is determined by rated capacity in tons, throat depth, and the number of punch stations in the turret.

Size	1	-	7-1/2	Tons,	18 Inch Throat Depth,	18 Station.
Size	2	-	12	Tons,	26 Inch Throat Depth,	20 Station.
Size	3	-	15	Tons,	26 Inch Throat Depth,	20 Station.
Size	4	-	25	Tons,	32 Inch Throat Depth,	24 Station.
Size	4A	-	30	Tons,	30 Inch Throat Depth,	12 Station.
Size	5	-	40	Tons,	32 Inch Throat Depth,	24 Station.
Size	6	-	50	Tons,	32 Inch Throat Depth,	22 Station.
Size	6A	-	50	Tons,	30 Inch Throat Depth,	12 Station.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSG, Memphis, TN 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3445

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Service.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

2.1.2 Other Government documents. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be that in effect on the date of the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATION, SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of documents, U.S. Government Printing Office, Washington , DC 20402-0001.)

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(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise Specified issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment document which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURER'S ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering - Modules and Diametral Pitches.

(Application for copies should be addressed to the International Organization for Standardization, c/o American National Standards Institute, ATTN: Scales Departments, 1430 Broadway, New York, NY 10018-3363).

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, 7901 Westpark Drive, McLean, VA 22101-4269.)

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Steet, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or distributed the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The punching machine shall be designed to accurately produce identical sheet metal parts from a template using punches and dies mounted in the machine's turret. The operator shall select the desired punch station, and actuate the machine for punching. Means shall be provided for accurately locating holes and cutouts on the workpiece. The machine shall be new (not a prototype) and one of the manufacturers' current production models capable of punching various shapes and sizes in sheet metal in accordance with the requirements herein. The machine design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, or repair.

3.2.1 Design, construction, and system of measurement. Unless otherwise specified, either the inch-pound system or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system shall be as specified (see 6.2.1). In this specification, all dimensions, measurements, sizes, and capacities are given in Inch-pound units. The dimensions may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

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3.2.2 Measuring and indicating device calibrations. Unless otherwise specified, either the inch-pound or the SI system of measurement shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 6.2.1), measuring and indicating devices shall be graduated to read in both the inch-pound and the SI system of measurements.

3.2.2.1 Dials. All dials used to indicate machine axes position or workpiece movement shall be graduated in Increments not larger than 0.001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal operating position.

3.2.3 Reclaimed materials. The punching machine may contain reclaimed materials provided such materials will not jeopardize the intended use, performance, and design life of the machine. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

3.2.4 Energy efficiency. The punching machine and its components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.5 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located conveniently to the operator's work station.

3.2.6 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all requirements of 29 CFR 1910. Additional safety and health requirements shall be as specified in detail (see 6.2.1).

3.2.6.1 Safety guarding. Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance, and repair of the parts. The safety devices shall not interfere with the operation of the machine. If specific point-of-operation guarding dependent on workpiece size and configuration is required, the procuring activity shall specify the exact configuration of the guard required (see 6.2.1).

3.2.7 Mercury restriction. The punching machine shall not contain mercury of mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the punching machine.

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3.2.9 ENVIRONMENTAL protection. The punching machine shall be so designed end constructed that, under the operation, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prohibited by Federal, state, or local statutes in effect at the point of installation .

3.2.10 Lubrication. Means shall be provided to ensure complete lubrication for all moving parts. When specified (see 6.2.1), an automatic lubricating system shall be provided for all parts and mechanisms requiring a continuous flow of lubricating oil. Means or determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or a similar device. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing, Each system shall include a cleanable or replaceable filter.

3.2.11 interchangeability All parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The punching machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale, and mismatching. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the machine. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer’s commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer’s standard commercial practice and color.

3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standards" section referenced therein.

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3.3.7 Gears. All gears of the ram and drive train shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the inch-pound system or IS0 54 for the metric (SI) system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened steel, Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirement shall be as specified.

3.4 Components. The punching machine shall consist of not less than a frame, turrets, punch holders, die holders, punches and dies, strippers, drive (mechanical or hydraulic), and electrical system.

3.4.1 Frame. The frame structure shall be either cast iron, cast steel, or a fabricated steel weldment of the mass, weight, and rigidty to withstand stresses produced when operating the machine at its maximum capacity without permanent deformation or failure. The frame shall be of "C" style configuration, stress relieved, with the overhang column and base forming a workpiece opening or throat. The depth of the throat shall be as specified for the machine sizes listed in Table I. The frame enclosure shall have covered access openings of the size required to facilitate servicing, maintenance, and adjustments of all components contained therein. The frame shall be a floor mounting unit, arranged to support the punching ram at a working height of not less than 36 inches, and not more than 42 inches from the mounting surface. The base of the frame shall have means to permit leveling and fastening the machine to the floor or foundation. When specified (see 6.2.1), vibration dampening leveling pads shall be provided. An eye or attachment point shall be provided at or near its balance point on the frame for lifting the machine. The eye or lift point shall be capable of supporting twice the weight of the machine, and its location shall not cause distortion to the frame when the machine is suspended,

3.4.2 Turrets. The upper and lower turrets shall be mounted on bearing surfaces between the overhang and base of the "C" type frame. The turrets shall be synchronized to rotate as a single unit with the punches of the upper turret constantly aligned with the corresponding dies of the lower turret. Unless otherwise specified (see 6.2.1), turret rotation shall be driven by a hydraulic or electric motor gear drive system. The machine shall be provided with indexing pins for aligning and locking the turrets into punching position under the punch ram. The mating holes of the indexing pins shall be provided with bushing inserts, The machine shall be provided with means for preventing the ram from being actuated while the turrets are not locked in the punching position.

3.4.2.1 Punch holders. Each punch station of the upper turret shall be provided with a punch holder. The punch holders shall be constructed of steel, and be arranged to securely hold the punch in position. The punch

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TABLE I. Sizes and capacities.

ITEM	Requirements							
	Size 1	Size 2	Size 3	Size 4	Size 4A	Size 5	Size 6	Size 6A
Rated capacity, tons	7-1/2	12	15	25	30	40	50	50
Number of punch stations	18	20	20	24	12	24	22	12
Throat depth, inches	18	26	26	32	30	32	32	30
Clearance between dies and strippers, inches	5/16	5/16	7/16	7/16	5/16	7/16	7/16	5/16
Length of ram stroke, inches	3/4	3/4	3/4	1	5/8	1	1	5/8
Hole punching capacity in 10 ga. (0.1345) thick mild steel, diameter, inches	3/4	1-1/8	1-3/8	2-3/8	2-3/4	3-3/4	4-1/2	4-1/2

NOTE: Requirements are "not less than" those shown for each machine size.

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holders shall be keyed to the turret stations to assure positive alignment with the punches and dies. The punch holders shall be designed to accommodate standard round and shaped hole punches.

3.4.2.2 Die holders. Each punch station of the lower turret shall be provided with a die holder or other suitable means for holding the dies in position The die holders shall be constructed of forged or stress relieves cast steel, and machined to securely hold the die in such a manner to prevent the die from turning in the turret. The die holders shall be designed to accommodate standard round and shaped hole dies.

3.4.2.3 Punches and dies. Unless otherwise specified (see 6.2.1), the machines shall be furnished with one set of punches and dies of sizes conforming to the requirements of Table II. The punches and dies shall be constructed of tool steel, hardened and ground for accuracy. The punches and dies shall be marked to identify each punch with the corresponding die When required additional punches and dies shall be furnished as specified (see 6.2.1).

3.4.2.4 Strippers. The machine shall be provided with strippers or other means for automatically releasing the workpiece from the punch when the punches are withdrawn from the dies.

3.4.3 Drive. Unless otherwise specified, the punching action of the machine shall be accomplished by either a mechanical or hydraulic type drive. When only one drive is acceptable, the particular type drive required shall be as specified (see 6.2.1). The drive shall be designed to assure that sufficient energy is developed for accomplishing the performance requirements specified herein. The machine shall provide means for adjusting both the depth of the ram stroke and stoke rate without stopping the machine. The machine shall include provisions for only one stroke of the ram for each depression of the foot pedal drive control.

3.4.3.1 Mechanical type drive system. The mechanical type drive system shall include an electric motor driven flywheel to transfer force to the punching ram. The drive shall convert rotary motion to reciprocating motion. The drive shall consist of an electric motor conforming to 3.4.4.1 and a belt drive arrangement powering a driven eccentric system. Rotating components shall be supported on bearing of sizes, types, and precision classes required for the specific applications to ensure maximum service life and accuracy. The drive shall be enclosed in and supported by the machine frame.

3.4.3.2 Hydraulic type drive system. The hydraulic type drive system shall conform to or exceed the requirements of JIC H-1 and shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Over pressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid The hydraulic reservoir shall have a means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable. The ram cylinder unit shall be provided with overload protection by a pressure relief mechanism or pressure switch on the ram.

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TABLE II. Punches and Dies.

Diameter of Punches Required (in inches)

Machine size 1	Machine size 2	Machine size 3	Machine size 4	Machine size 4A	Machine size 5	Machine size 6	Machine size 6A
1/8	1/8	1/8	1/8	1/8	1/8	1/8	1/4
5/32	5/32	3/16	5/32	1/4	5/32	3/16	3/8
3/16	3/16	1/4	3/16	3/8	3/16	1/4	1/2
7/32	1/4	5/16	1/4	1/2	1/4	5/16	3/4
1/4	5/16	3/8	5/16	3/4	5/16	3/8	1
9/32	11/32	7/16	11/32	1	11/32	1/2	1-1/2
5/16	3/8	1/2	3/8	1-1/4	3/8	5/8	2
3/8	7/16	9/16	7/16	1-1/2	7/16	3/4	2-1/2
7/16	1/2	5/8	1/2	1-3/4	1/2	7/8	3
1/2	9/16	11/16	9/16	2	9/16	1	3-1/2
5/8	5/8	3/4	5/8	2-1/2	5/8	1-1/4	4
3/4	11/16	15/16	11/16	3	3/4	1-1/2	4-1/2
7/8	3/4	1	3/4		7/8	1-3/4	
15/16	1	1-1/4	7/8		1	2	
1	1-1/4	1-1/2	1		1-1/4	2-1/4	
1-1/8	1-1/2	1-3/4	1-1/4		1-1/2	2-1/2	
1-1/4	1-3/4	2	1-1/2		1-3/4	2-3/4	
1-1/2	2	2-1/4	1-5/8		2	3	
	2-1/4	2-1/2	1-3/4		2-1/4	3-1/4	
	2-1/2	3	2		2-1/2	3-1/2	
			2-1/4		2-3/4	4	
			2-1/2		3	4-1/2	
			3		3-1/4		
			3-1/4		3-1/2		

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3.4.4 Electrical system. The electrical system shall con form to NFPA 79. Unless otherwise specified (see 6.2.1), the machine shall be wired to draw all of its electrical power from a single 230/460 volt, 3-phase, 60 Hz circuit. Unless otherwise specified (see 6.2.1), the machine shall be initially wired to operate from d 460 volt source.

3.4.4.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed, balanced, and permanently lubricated type. Unless otherwise specified (see 6.2.1), alternating-current (AC) motors shall be designed to operate on 60-HZ, and each motor enclosure shall meet the requirements for a drip-proof enclosure. All motcrs shall meet the requirements of NEMA MG-1.

3.4.4.2 Solid state components. Solid state design shall be used tnroughout for electronic components.

3.4.4.3 Control circuit voltage. The control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage no greater than 120 volts.

3.4.4.4 Worklight. When specified (see 6.2.1), an incandescent worklight shall be provided. The fixture shall be permanently mounted to the machine with an air vented protective reflector shield, on-off switch, and shall be adjustable to reflect maximum light on the work area. The lamp shall be no less than 75 watts.

3.4.5 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the machine. The meter shall be of the nonresetting type and shall have a range of O to 99,999 hours in increments of not greater than one hour, The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine. Upon reaching 99,999 hours, the meter readout shall automatically return to zero and continue to accumulate time.

3.4.6 Drop latch gauge. When specified (see 6.2.1), the size 2, 3, 4, 4A, 5, 6, and 6A machines shall be furnished with a drop latch gauge for aligning the workpiece under the ram and punch. The gauge shall be provided with a guide bar for aligning the workpiece along both the “X” and “Y” axes. The guide bar shall be of sufficient size and strength, permanently marked in graduated increments for measuring the distance from the center of the punch and ram to at least 48 inches along both the + X and - X axes. The guide bar shall be provided with at least 12 drop latch workpiece stops arranged to slide along and be clamped to the bar at intervals for hole locations. The guide bar shall be attached to a back gauge for controlling workpiece adjustment and alignment along the “Y” axis. The back gauge shall be a hand operated measurement counter to automatically indicate the distance between the guide bar and center of the ram and punch. The back gauge shall provide not less than 24 inches of workpiece adjustment. The drop latch gauge shall be provided with a work table arranged to enclose the turrets on the operator’s right and left sides and have total length of not less than 72”.

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3.4.7 Hole pattern duplicating attachment. When specified (see 6.2.1.), the machine shall be furnished with a hole pattern duplicating attachment. The attachment shall be provided with metal clamps, or other means, for securing the hole pattern to metal blanks. The attachment shall be arranged to permit the pattern and blank to be positioned under the ram and punch for allowing the punch to be lowered to enter the holes of the pattern. The attachment shall be provided with an adjustable limit switch to automatically actuate the ram to punch the blanks when the punch touches the blank. The attachment shall be adjustable to punch blanks up to 1/4 inch thick while accommodating patterns from 3/32 to 1/8 inch thick.

3.4.8 Stylus type workpiece duplicator. When specified (see 6.2.1), the machine shall be furnished With a stylus type workpiece duplicating gauge for aligning the workpiece under the 'ram and' punch. The gauge shall be a pantograph styled arrangement with a template table, workpiece holding carriage, and hand controlled stylus for locating the punching points of the templates and actuating the machine drive for punching. The work table shall have provisions for holding templates up to the sizes-specified in table III. The carriage shall be provided with at least two workpiece c1amps. Means shall be provided for selecting the desired punch station and locking the turrets into punching position from the operator's position.

3.4.8.1 Digital readout (DRO) system. When specified (see 6.2.1), a DRO system shall be provided on the Y and X axes of travel of the duplicating attachment. The Y and X axes of travel of DRO system shall be not less than the duplicating area specified in table 1. The DRO system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to assure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.001 inch. The system accuracy shall be 0.001 inch per foot and the repeatability shall be 0.001 inch with a non-accumulative tolerance of 0.005 inch over the full duplicating range of travel . Special requirements for the DRO system components shall be as specified (see 6.2.1). The DRO shall indicate axis location in both the US and SI units. The DRO system shall be provided with electronic and manual calibration to the machine on which it is installed to eliminate the effects of distortion error. An input power source fluctuation of ± 10 percent shall not adversely affect the DRO system operation.

3.4.8.2 Position feedback devices. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the DRO unit. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback devices. The connecting cable from the position feedback device shall be electrically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall De made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related

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TABLE III. Duplicator - sizes and capacities.

Machine Size	1	2	3	4	4A	5	6	6A
Duplicating Area	12 X 18	18 X 24	24 X 26	26 X 36	30 X 54	26 X 72	26 X 72	30 X 54
Sheet Size	16 X 18	22 X 24	28 X 26	26 X 45	36 X 60	26 X 81	26 X 81	36 X 60

NOTE: All dimensions are specified in inches.

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components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines in any multiple machine order.

3.4.8.3 Electronic display console. The electronic display console for the DRO system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wise viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's work position. The console shall be mounted to withstand shock and vibration generated by the machine. The connecting cable shall be armored and shall be installed so as not to interfere with the normal operations of the machine. The cable shall be securely fastened to the structure of the machine in accordance with the requirements of the contract.

3.5 Size and capacity. Unless otherwise specified (see 6.2.1), the size and capacity of the machine shall be not less than those specified in table I for the machine size specified.

3.6 Performance. The machine shall be capable of performing sheet metal punching operations in the workpiece thickness specified in table I for the size machine required. The machine shall be capable of performing turret punching operations in workplaces of mild steel having "not less than" 50,000 PSI tensile strength without permanently deforming the workpiece.

3.6.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2.1).

3.7 Standard equipment. Unless otherwise specified, all standard equipment normally furnished with the manufacturer's commercial machine shall be furnished. When specified (see 6.2.1), required additional equipment shall be furnished and fully described.

3.8 Optional equipment. Optional equipment shall be furnished as specified and shall be fully described (see 6.2.1).

3.9 Repair parts. Repair parts shall be furnished as specified (see 6.2.1).

3.10 Marking on plates and charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a Contrasting background. All plates shall be corrosion resistant.

3.10.1 Marking on plates, charts, and decals. All words on plates, charts, and decals shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practical, the instructions shall be applied directly

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onto the item by environmentally protected decalcomania transfer, a metal wrap-around tag, stenciling, silk screening, or by any other method suitable for the use intended. Marking shall be as permanent as the normal life expectancy of the item to which they are applied and shall be capable of withstanding the environmental test and cleaning procedures specified for the item.

3.10.2 Lubrication plate. A corrosion resistant lubrication plate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.10.3 Nameplate. A corrosion resistant nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the machine and accessories shall meet all requirements specified herein and shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspections system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2.1) all inspections, tests, and examinations shall be performed in an indoor facility with ambient conditions of 65 to 90 degrees Fahrenheit and 20 percent to 90 percent relative humidity.

4.4 First article inspection. When first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests shall be cause for rejection.

4.5 Quality conformance inspection. A quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), the quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, all tests, and the inspection shall be cause for rejection.

4.6 Examination. The machine shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance tests. The machine's performance shall be tested by operating the machine to perform sheet metal punching operations to determine conformance with the requirements of 3.6 and Table I. The machine shall punch not less than six holes each of the diameters specified in table II and in the material thickness and type as specified in table I.

4.7.3 Electromagnetic interference control tests. Equipment requiring electromagnetic Interference control shall be tested to determine conformance with 3.3.8 using the procedures given in MIL-STD-462.

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4.8 Packaging inspection. Packaging of each item shall be inspected to determine conformance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The machines covered by this specification are intended to incorporate simple design, rapid tool change, and easy tool adjustment for use in prototype or short run sheet metal shearing, nibbling, and forming operations. Different operations may be performed on the same machine without expensive tooling and movement of job to other machines.

6.2 ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size machine required (see 1.2 and Table I).

c. First article, if required (see 3.1).

d. If machine design and construction is required in a specific measurement system (Inch-pound or SI), state required system (see 3.2.1).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (inch-pound or SI), state required system (see 3.2.2).

f. Dual calibrations (inch-pound and SI), if required (see 3.2.2).

g. Additional safety and health requirements, if required (see 3.2.6).

h. Specify configuration of additional point-of-operation guarding, if required (3.2.6.1).

i. Automatic lubrication, if required (see 3.2.10).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

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l. Vibration dampening leveling pads, if required (3.4.1).

m. If turret drive is to be a specific type, state required type (see 3.4.2).

n. Specify and fully describe punches and dies, if different (see 3.4.2.3).

o. Specify and fully describe additional punches and dies, if required (sede 3.4.2.3).

P. If machine drive is to be a specific type, state required type (see 3.4.3).

q. Electrical system, if different (see 3.4.4).

r. Initial wiring, if different (see 3.4.4).

c, Motor, if different [see 3.4.4.1).

t. Work ight, if required (see 3.4.4.4).

u. Hour meter, if required (see 3.4.5).

v. Drop latch gauge, if required (see 3.4.6).

w. Hole pattern duplicating attachment, if required (see 3.4.7).

x. Stylus type workpiece duplicator, if required (see 3.4.8).

y. Digital readout system, if required (see 3.4.8.1).

z. Specify any special DRO system requirements, if required (see 3.4.8.1).

aa. Size and capacity, if different (see 3.5).

bb. Additional performance features, if required (see 3.6.1).

cc. Standard equipment, if different (see 3.7).

dd. Specify and fully describe optional equipment, if required (see 3.8).

ee. Specify applicable repair parts, if required (see 3.9).

ff . Lubrication plate, if different (see 3.10.2).

gg. Nameplate, if different (see 3.10.3).

hh. Inspection conditions, if different (see 4.3).

ll. First article inspection, if different (see 4.4).

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j). Quality conformance inspection, if different (see 4.5).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports should be specified on a OD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First Article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract, The contracting officer should include specific instructons in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Cross-reference of classification changes. The machine classifications are the same as the previous issue of this specification with the exception of the addition of sizes 4A. 6 and 6A which were not covered in MIL-P-80026B. Table 111, duplicator-sizes and capacity were added to the specification.

6.5 Subject term (key word) listing.

Punching

Turret

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety health requirements. 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors. air contaminants and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity is advised to analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.8 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.9 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Preparing activity:
DLA-IP

(DOD Project 3445-0111)

Review activity:
Air Force - 84

User activity:
Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-80026C	Punching Machine, Metal, Power Driven, Turret Type

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording.

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 87 MAR 1423 PREVIOUS EDITION IS OBSOLETE.

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